

choptuik_ac_bc · theorems12 · github.com/wild8highlander/choptuik_ac_bc

The Weight-Rule Theorem: $(\lambda + p) \dots (\lambda + p + n - 1)F$ and the invariance of the cubic pencil

12/12 cases exact · $\text{char3} = (1 + \lambda^2)^9 \cdot \text{char2} \cdot \pm i$ are phantoms (40 digits)

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Abstract

This monograph records the weight-rule theorem of the third order of the truncated tower: for all pairs $(p, n) \in \{0, 1, 2, 4\} \times \{1, 2, 3\}$ the n -th-order table of the profile residue of weight p factorizes exactly as the product $(\lambda + p)(\lambda + p + 1) \dots (\lambda + p + n - 1)F$ with constant coefficients, a zero completeness residual and the leading coefficient 1 at all second-derivative slots — a coefficient previously accepted by precedent is derived here for the first time.

The main corollary is prolongation invariance: the cubic pencil of the prolonged system satisfies the exact identity $M3^T M3 = (1 + \lambda^2)M2^T M2$, the characteristic polynomials are linked by the law $\text{char}3 = (1 + \lambda^2)^9 \cdot \text{char}2$, the rank drops are unchanged, and the pair $\pm i$ is refuted by the 40-digit filter as a phantom. The third order adds no genuine linear dynamics; $\delta_{\text{mono}} = 0$ is stable. All rule polynomials, the tables of layers T1–T4, the corollaries, the honest caveats and the reproduction protocol are included.

Keywords: weight rule, higher-order tables, cubic pencil, prolongation invariance, phantoms $\pm i$, SymPy, machine proof.

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1. Introduction

The fourth monograph records the weight-rule theorem — the exact law by which the higher-derivative tables of the truncated tower factorize through the senior profile factor — together with its main corollary: the prolongation invariance of the cubic pencil. The result was delivered by campaign v14 (machine `sympy_third_order.py`, verdict `third_order_tables.json`, LEDGER §25) as the answer to the question: does the de-adiabatic pattern $(l + 2)(l + 3)F / (l + 4)(l + 5)F$ of the second order extend to the third order, and does the third order add genuine linear dynamics?

The answer is twofold. Yes, the pattern extends: all third-order tables factorize as the exact product $(\lambda + p)(\lambda + p + 1)\dots(\lambda + p + n - 1)F$ for all twelve pairs (p, n) of the studied set, and the coefficient 1 at the second derivatives — the one accepted “by the ddQ precedent” in machine v13 — is derived for the first time, not chosen. And no, nothing new appears: the cubic pencil of the prolonged system $\{F = 0, dF/d\tau = 0\}$ possesses the exact identity $M3^T M3 = (1 + \lambda^2)M2^T M2$, the characteristic polynomial factorizes as $\text{char3} = (1 + \lambda^2)^9 \cdot \text{char2}$, and the pair $\pm i$ are phantoms refuted by the 40-digit filter. The spectrum is prolongation-invariant; the third order adds no genuine linear dynamics.

2. Notation and Setting

A profile of weight p is a tower amplitude entering the sources with the weight $s^{(+p)}$, where s is the scale factor of the adiabatic representation. An n -th-order table is the n -th derivative with respect to the log-time y of the profile residue: its right-hand side is expressed through the amplitudes and their derivatives up to order n . The pencil $M(\lambda)$ of the linearized prolonged system is a matrix pencil whose rank drops determine the spectrum; $M2$ is the second-order pencil (quadratic in λ), $M3$ the cubic one built from the prolongation $\{F = 0, dF/d\tau = 0\}$.

The setting: to prove for all pairs $(p, n) \in \{0, 1, 2, 4\} \times \{1, 2, 3\}$ that the n -th-order table of the weight- p profile equals the product $(\lambda + p)\dots(\lambda + p + n - 1)F$ with constant coefficients and leading coefficient 1 at every second-derivative slot; then to prove the exact pencil identity and its consequence for the characteristic polynomials.

Table 4.1. Notation of the monograph

Symbol	Meaning
p	weight of the profile amplitude in the sources (s^p)
n	order of the table (n -th derivative in y)
F	the profile residue (closing slot of the second derivatives)
λ	spectral parameter of the pencil
$M2, M3$	quadratic and cubic pencils of the prolonged system
$\text{char2}, \text{char3}$	characteristic polynomials of the pencils $M2, M3$

3. Statement of the Theorem

Theorem 4 (the weight rule and prolongation invariance). For all pairs $(p, n) \in \{0, 1, 2, 4\} \times \{1, 2, 3\}$:

(i) the n -th-order table of the weight- p profile factorizes exactly:

$$T_n[p] = (\lambda + p)(\lambda + p + 1) \dots (\lambda + p + n - 1) \cdot F \quad (4.1)$$

with constant coefficients (`coefficients_constant = true`), zero completeness residual (`completeness_residual = 0`) and leading coefficient 1 at every second-derivative slot (`leading_coeff_1 = true`);

(ii) the third-order tables are exact y -derivatives of the second-order ones: $W2'' = (6W2h + 5dW2h + ddW2h)/s^4$, $W2''' = (24W2h + 26dW2h + 9ddW2h + dddW2h)/s^5$ and analogously for $R3, P4, R5$; the mode factorizations are: $W2/R3 \text{ --- } \lambda^3 + 9\lambda^2 + 26\lambda + 24$, $P4 \text{ --- } \lambda^3 + 15\lambda^2 + 74\lambda + 120$, $R5 \text{ --- } \lambda^2 + 9\lambda + 20$;

(iii) the cubic pencil of the prolongation is structurally invariant: $M3 = [M2; \lambda M2]$ and

$$M3^T M3 = (1 + \lambda^2) \cdot M2^T M2 \quad (4.2)$$

(iv) the characteristic polynomials are linked by the exact law $\text{char3} = (1 + \lambda^2)^9 \cdot \text{char2}$; the rank drops of $M3$ coincide with those of $M2$; the pair $\pm i$ is a phantom (relative $\sigma_{\min} =$

2.285×10^{-5} on the 40-digit filter), the multiplicity of $\lambda = 0$ is unchanged (the λ^4 structure of session 17).

4. Sketch of the Machine Proof

4.1. The twelve cases of the weight rule

The machine `sympy_third_order.py` builds the tables symbolically: each y -derivative is computed by the weight rule $d/dy = (1/s)(d/d\tau + \text{weight})$, after which the coefficients at every slot are compared with the product $(\lambda + p) \dots (\lambda + p + n - 1)F$. All twelve cases are closed with a zero completeness residual; the rule polynomials are written in Table 4.2. Essentially, the coefficient 1 at the dd -slots is now derived: it is inherited from the factorization, not postulated as in `v13`.

Table 4.2. The weight-rule polynomials (all 12 cases exact, residuals = 0)

p	n = 1	n = 2	n = 3
0	λ	$\lambda^2 + \lambda$	$\lambda^3 + 3\lambda^2 + 2\lambda$
1	$\lambda + 1$	$\lambda^2 + 3\lambda + 2$	$\lambda^3 + 6\lambda^2 + 11\lambda + 6$
2	$\lambda + 2$	$\lambda^2 + 5\lambda + 6$	$\lambda^3 + 9\lambda^2 + 26\lambda + 24$
4	$\lambda + 4$	$\lambda^2 + 9\lambda + 20$	$\lambda^3 + 15\lambda^2 + 74\lambda + 120$

4.2. The pencil and its invariance

The prolongation of the system $\{F = 0\}$ by the τ -derivative adds the equation $dF/d\tau = 0$; the corresponding cubic pencil is built as $M3 = [M2; \lambda M2]$. The machine verifies the structural identity (4.2) exactly (`structural_identity_small_exact = true`) and at three rational control points $\lambda = 1/2, 1, 2$ (`det_sanity_exact_rational_l: all true`). The consequence — the factorization $\text{char3} = (1 + \lambda^2)^9 \cdot \text{char2}$ — means that the only new roots of the cubic pencil are $\lambda = \pm i$, each ninefold. The numerical filter on 40 digits: $\sigma_{\min}/\|M\| = 2.285 \times 10^{-5}$ for $\lambda = \pm i$ (PHANTOM) versus 0.0 for $\lambda = 0$ (GENUINE).

Table 4.3. Layers of the verdict third_order_tables.json

Layer	Statement	Status
T1_weight_rule	rule (4.1) exact for 12/12 pairs (p, n); leading coeff = 1 derived	all_exact = true
T2_tables	third tables = exact y- derivatives of the second; mode factorizations hold	all_exact = true
T3_regression	leading coefficient = 1 in all generations; dd = 0 $\Rightarrow$ exact adiabatics	all_ok = true
T4_cubic_pencil	$M3^T M3 = (1+\lambda^2)M2^T M2$; char3 $= (1+\lambda^2)^9 \text{char2}$; $\pm i$ are phantoms	exact + 40 digits

5. Numerical Results

The regression layer T3 links the generations of tables: the leading coefficient of the slot equals 1 in the third order (third_W2_dddd = 1), in the second (second_W2_dd = 1, second_P4_dd = 1) and in the ddQ precedent of machine v13 (machine_ddQ = 1); when the second derivatives are zeroed, the second-order tables pass into the adiabatic ones exactly (second_to_adiabatic = 0). The de-adiabatic pattern of v13 (the structures 5,6 = $(\lambda + 2)(\lambda + 3)$ and 9,20 = $(\lambda + 4)(\lambda + 5)$) thus turns out to be special cases of the general rule (4.1).

Table 4.4. The phantom filter (40 digits, layer T4)

λ	$\sigma_{\min} / \ M\ $ (rel, 40d)	Verdict
+i	2.285×10^{-5}	PHANTOM
-i	2.285×10^{-5}	PHANTOM
0	0.0	GENUINE

Jointly, Tables 4.2–4.4 close the question: the third order adds no genuine linear dynamics. The spectral structure {0-multiplicity 4, phantom pairs} is unchanged; the phantom gap to

the $\pi/30$ ladder ($\sim 2.9\%$) remains outside linear statics and requires finite-amplitude mechanisms (campaigns v13–v15, monographs No. 5–No. 6).

6. Corollaries

Corollary 4.1 (a derived, not chosen law). The coefficient 1 at the dd-slots — the key assumption of the de-adiabatization of v13 — is now a consequence of the factorization (4.1). The independent dd-law receives an algebraic justification, which prepares the dd-closure theorem (monograph No. 5): the y -prolongation coincides with the τ -prolongation precisely because the table derivatives factorize by the weight rule.

Corollary 4.2 (the stability of $\delta_{\text{mono}} = 0$). Since $\text{char3} = (1 + \lambda^2)^9 \text{char2}$ and $\pm i$ are phantoms, the v13–v14 conclusion of a zero linear monodromy shift is stable against the third order: a stronger prolongation produces no new genuine modes, and the residual $\sim 2.9\%$ to $\pi/30$ cannot be produced by a linear sector of any order from the considered class.

Corollary 4.3 (prolongation invariance as a protocol). Identity (4.2) provides a cheap control protocol for any future enhancement: if a prolonged pencil is built correctly, its singular drops must coincide with the drops of the original. A violation of this identity in any future machine is a signal of a construction error, not of physics.

7. Honest Caveats

The rule is proved on the set $\{0, 1, 2, 4\} \times \{1, 2, 3\}$; the extrapolation to arbitrary (p, n) is plausible (the product structure is visible) but formally remains a hypothesis until a separate check. The pencil M3 is built for the linear prolongation $\{F = 0, dF/d\tau = 0\}$; the nonlinear prolongation with x -dependent Jacobians (campaign v15) goes beyond the present theorem and is studied in monograph No. 6. The phantom verdict rests on a 40-digit threshold filter: like every numerical filter it is calibrated and may err on structures with exceptionally small σ_{min} ; for $\pm i$ the margin (10^{-5} against the machine zero of $\lambda = 0$) is comfortable. Finally, the factor $(1 + \lambda^2)^9$ reflects the dimension of the concrete system; under a tower enhancement the exponent changes and the identity must be rechecked — the protocol of Corollary 4.3 exists exactly for that.

8. Reproducibility

Table 4.5. Reproduction protocol for Theorem 4

Step	Command	Result
1	<code>python3 sympy_third_order.py</code>	<code>results/third_order_tables.json</code> (layers T1–T4)
2	<code>python3 sympy_dd_closure.py</code>	<code>results/dd_closure.json</code> (justification of the dd-law)
3	<code>python3 -m pytest tests/ -q</code>	58 passed (the weight-rule tests included)

The machine running time is 11.4 s (`runtime_s` of the verdict) on an ordinary laptop; the machines are single-threaded and deterministic. The data correspond to commit 5b256cd; an autonomous copy of the machine lives in `einstein_direct/open9/machines/` with a private copy of the results.

References

- [1] Repository `wild8highlander/choptuik_ac_bc`: `einstein_direct/sympy_third_order.py`; `einstein_direct/results/third_order_tables.json` (campaign v14, LEDGER §25).
- [2] Ibid.: `einstein_direct/sympy_dd_closure.py`; `einstein_direct/results/dd_closure.json` (session 18b).
- [3] Ibid.: `einstein_direct/sympy_second_flows.py`; `einstein_direct/results/second_flows_exact.json` (campaign v13).
- [4] Ibid.: `einstein_direct/README.md` §§1.1, 3.6; `einstein_direct/README_EN.md` §25.
- [5] Monographs No. 1, No. 5, No. 6 of the theorems12 cycle.